

SUMEDHA UMESH KULKARNI

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PROFILE SUMMARY

Software Engineer with experience in Embedded Systems (C++, Bluetooth, RTOS), Full-Stack Development (MERN, REST APIs, GCP), and AI/ML (RAG pipelines, OCR, NLP). Authorized to work in the US on F1-OPT with STEM extension, open to relocation.

EDUCATION

University of Southern California	Los Angeles
Master of Science in Computer Science, CGPA:3.74	August 2024-May 2026
• Coursework: Analysis of Algorithms, Web Technologies, Machine learning, Database systems, Natural language processing	
KLS Gogte Institute of Technology	India
Bachelor of Technology in Computer Science and Engineering	July 2019-August 2023

TECHNICAL SKILLS

- Programming Languages: Python, C++ , Java, JavaScript, SQL
- Databases and Operating system: MySQL, PostgreSQL, MongoDB, Linux, Windows, MacOS
- Frameworks & Libraries: Django, React, Angular, Node.js, SwiftUI, NumPy, Pandas, Sci-kit Learn, TensorFlow, Keras, OpenCV
- Cloud Platforms: Google Cloud Platform (GCP), Microsoft Azure
- Tools & Technologies: Git, Docker, GTest, GDB, CMake, QEMU, Valgrind

PROFESSIONAL EXPERIENCE

Deep Cognition	Remote
AI intern	June 2025-August 2025
• Built a Haystack-based RAG pipeline to split/index Markdown, embed with SentenceTransformer, and query Anthropic Claude, cutting retrieval latency by 37%	
• Solved real-time Q&A across 500+ documentation repositories in LibreChat by exposing the RAG tool through a Flask server with RESTful JSON endpoints	
• Accelerated invoice ingestion by 65% by automating EML-to-PDF conversion with Playwright, extracting metadata from PDFs/images via OCR, and queueing 200+ documents daily to AI-driven document processing system	
Mercedes Benz	India
Software Developer	August 2023-May 2024
• Designed a C++17 solution for HAL-to-Application communication using BlueGo and DeviceManager RPC to retrieve/persist battery data and deliver real-time UI alerts, improving responsiveness and system reliability	
• Analyzed and fixed Bluetooth and DeviceManager related bugs using GDB and DLT-Viewer, addressing core dump issues and leading to a 20% reduction in overall bug incidence	
• Increased code coverage by 30% and enhanced system reliability by automating unit testing with GTest	
• Led hardware bring-up and flashing activities for 45+ development tasks using ET-framework and Monaco tools	
Intern	February 2023-July 2023
• Created Python command-line application in Linux for automated hardware and software testing, decreasing manual effort, and adopted by 10+ engineers for daily workflows	

ACADEMIC PROJECTS

Deep Research Agent LangGraph, Python, Tavily, Anthropic Claude
• Deployed Built a LangGraph-based multi-agent system that decomposes open-ended queries, orchestrates parallel research workflows across 500+ sources, and generates citation-backed reports, reducing research turnaround time by ~60%
Concierge: Proactive On-Device In-Car AI Agent Python, FastAPI, llama.cpp, WebSockets, Whisper
• Built a two-stage gate-and-generator loop around a 1.2B-parameter on-device LLM with GBNF grammar-constrained JSON, eliminating parse failures and capping LLM calls at 10% of 3-second tick
• Crafted an intent decomposition engine and tool router that parses compound voice commands into typed sub-intents, dispatching to 4 keyless APIs (Overpass, Open-Meteo, OSRM, Nominatim) under 1 second
Embedded Health-Monitor Firmware Zephyr RTOS, C++17, Cortex-M, QEMU, GoogleTest
• Implemented C++17 firmware on Zephyr RTOS for ARM Cortex-M with a 3-tier layered architecture (drivers, services, algorithms), validated in QEMU under CI
• Integrated Programmed a 2-stream signal pipeline covering heart-rate (IIR bandpass + peak detection → BPM) and activity (IMU magnitude variance), verified by host-side GoogleTest suites for fast iteration without target flashing